ABHINAV KUMAR

Full Stack & Backend Engineer · Distributed Systems · Applied AI (RAG, Embeddings)

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EDUCATION

B.Tech in Computer Science, Specialization in Full Stack AI, UPES Dehradun *(CGPA: 8.29, 100% merit scholarship)* 2022 – 2026

Intermediate (Class XII), CBSE *(84%)* 2019 – 2021

EXPERIENCE

Application Engineering Intern – Innovaccer *(Healthcare Data Platform)* [Certificate] Jan 2026 – Aug 2026

- Resolved **~30 production bugs** across **FastAPI backend services** and the React frontend, tracing each to root cause and shipping tested fixes.
- Enhanced **ProfileSyncJob** to fix sync drift between **MongoDB and Elasticsearch**, improving consistency of patient data served by search.
- Delivered features for **Rule Engine, Consent Management, Patient Search and Clinical Cards**, core clinical workflows of the platform.
- Worked in an **Agile/Scrum** pod (sprint planning, refinement, standups, Kanban, retros) alongside QA and product.

Full Stack Development Intern – Xebia *(Project: Learnify)* Jun 2025 – Jul 2025

- Built the first version of **Learnify**, a **MERN** ed-tech platform with student and instructor workflows, **JWT authentication** and role-based access control.
- Integrated **Razorpay** checkout, **Cloudinary** media management and a **Gemini-powered chatbot** for learner questions.

PROJECTS

Learnify – Production-Grade E-Learning Platform | *MERN, Redis, RabbitMQ, Elasticsearch, Qdrant, Docker, Prometheus* 2025 – Present

- Evolved the internship project into an **event-driven** system (API, worker, AI microservice) using a **transactional outbox** + **RabbitMQ** with retry/dead-letter queues, idempotent consumers and a refund saga.
- Added a **Redis** cache shared across instances with versioned invalidation and stampede protection: **8.5x throughput** (141 → 1,207 req/s), avg latency **352 → 41 ms**.
- Moved search to **Elasticsearch** (typo-tolerant, autocomplete, facets, zero-downtime reindex), cutting search p95 from **4.33 s to 76 ms** at **4x** throughput.
- Built a **RAG study assistant** (**Gemini** embeddings, **Qdrant**, per-course retrieval, cited answers), tuned on an eval set to **12/12 hit@6** with all off-topic queries rejected.
- Secured with **rotating refresh tokens** (reuse detection), **TOTP 2FA**, RBAC, an append-only audit log, **idempotency keys** on payments and **HMAC-signed webhooks**.
- **k6** load tests exposed an unpaginated catalog and slow highlighting; fixes cut journey **p95 from 8.3 s to 25 ms**, and covering **MongoDB indexes** cut a 200K-doc scan to 12.
- Instrumented with **Prometheus, Grafana, OpenTelemetry** (traces across RabbitMQ); **150+ Jest tests** and **GitHub Actions** CI publishing Docker images.

Sense-Cluster – Semantic Q&A Platform | *FastAPI, PostgreSQL, pgvector, React, Sentence-Transformers* Jan 2025 – Present

- Prototyped duplicate-question detection with **BERT embeddings** and cosine similarity on **Quora Question Pairs** (300K+ pairs), reaching **80% accuracy** and beating a **TF-IDF** baseline.
- Rebuilt as a full-stack **Q&A app**: questions are embedded with **all-mpnet-base-v2** and matched by **pgvector** cosine search over an **IVFFlat index** with a tunable threshold, flagging duplicates before they are posted.
- Implemented **RBAC** (admin, contributor, member) with an approval workflow, **JWT + Argon2** auth and **rate limiting**; backed by **24 pytest** tests, **Alembic** migrations and **GitHub Actions** CI (pytest, ESLint, build).

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, C++, SQL
Backend & APIs	Node.js, Express.js, FastAPI, REST, Socket.IO, OpenAPI, SQLAlchemy, Mongoose
Frontend	React.js, Redux Toolkit, Tailwind CSS, HTML, CSS, Vite
Data & Messaging	MongoDB, PostgreSQL, Redis, Elasticsearch, RabbitMQ, Qdrant, pgvector
AI / ML	RAG, Embeddings, Sentence-Transformers, BERT, Google Gemini, Semantic Search
DevOps & Testing	Docker, Kubernetes, GitHub Actions (CI/CD), Prometheus, Grafana, OpenTelemetry, Jest, pytest, k6, Git, Postman

ACHIEVEMENTS

- Solved **1000+ coding problems** across platforms; **1200+ Codeforces rating** (Pupil).
- Awarded a **100% merit-based scholarship** at UPES for ranking among the **top 50 students**.